Big Mountain

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What opportunities exist for Big Mountain Resort to increase profit in the WInter of 2022 by 10% from last year by increasing the traffic of skiers and riders with the new premium.

- **Context**
  - Location: Montana
  - 350,000 people ski or snowboard yearly
  - Current pricing strategy has been to charge a premium above the average price of resorts in its market segment
- **Criteria for Success**
  - QC System will be adopted and implemented for business use no later than the winter season of 2022
- **Scope of Solution Space**
  - Calculating and increasing the cost of premium so that it increases profit by 10%

- **Constraints with Solution Space**
  - Additional chair increases operating costs by $1,540,000 this season
  - Competition from other ski resorts
  - Not increasing the price so much that it deters and lowers the traffic
- **Stakeholders**
  - Director of Operations - Jimmy Blackburn
  - Database Manager - Alesha Eisen
- **Key Data Sources**
  - Clean up the csv file so that it focuses on the region of the Northern Rockies and Plains in the northwestern United States
  - Calculate the average cost of weekend of the selected areas

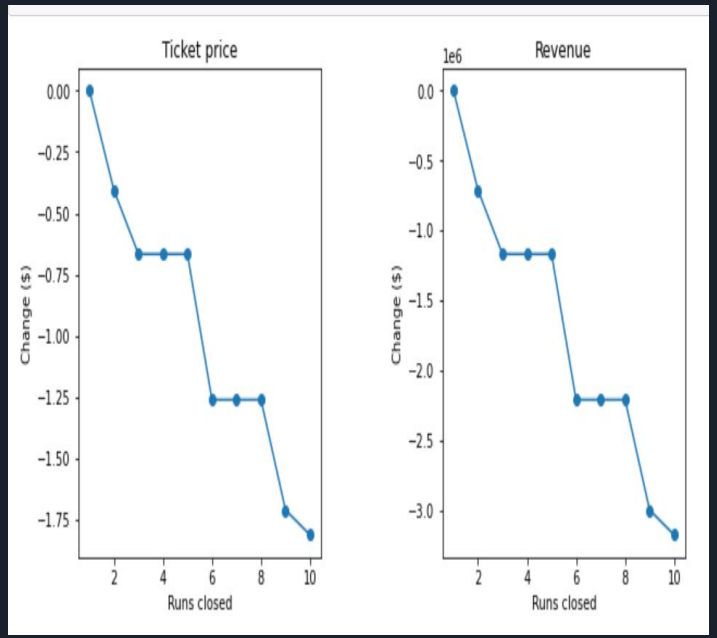
Recommendation and Key Findings

- **Recommendation**
  - Resort_night_skiing_state_ratio seems the most correlated with ticket price
    - Provide more night skiing
  - Guaranteed snow would cost in terms of snow making equipment, which would drive prices and costs up
  - The vertical drop seems to be a selling point that raises ticket prices as well

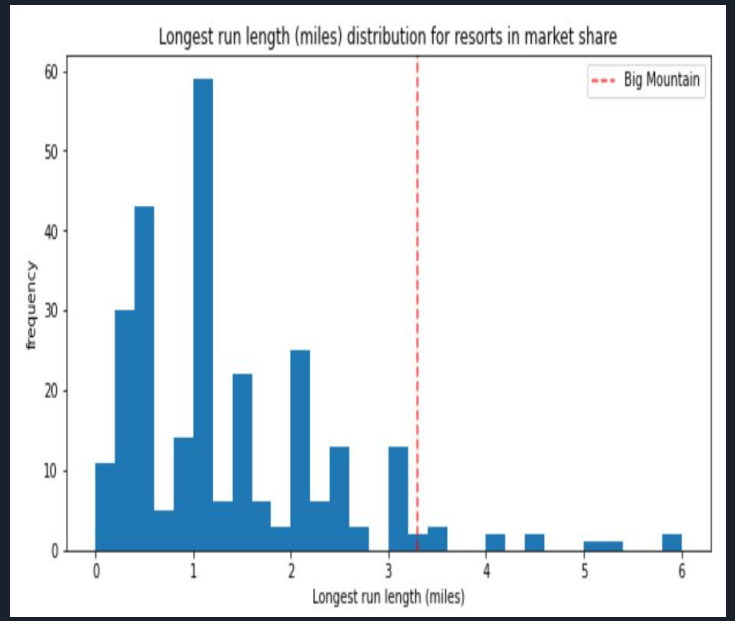
- **Key Findings**
  - It seems that when resorts are more densely located with population, more night skiing is provided
  - Visitors would seem to value more guaranteed snow,
  - AdultWeekend ticket price are correlated with fastQuads, Runs, Snow Making_ac

Modeling Results and Analysis

- Closing one run makes no difference
- Closing 2 and 3 reduces support for ticket price and so revenue
- Closing down 3 runs, it seems they may as well close down 4 or 5 as there's no further loss in ticket price
- Increasing the closures down to 6 or more leads to a large drop



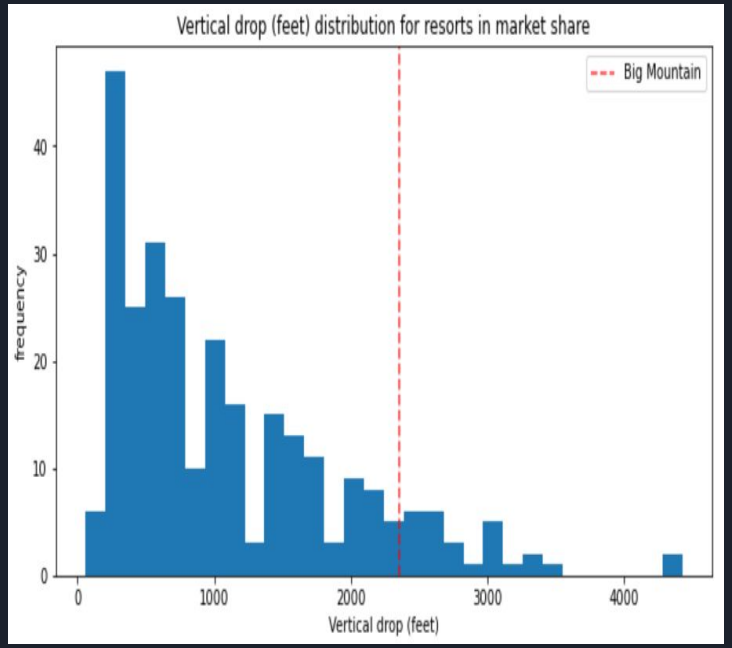
Longest Run Length (mi) Distribution for Resorts in Market Share



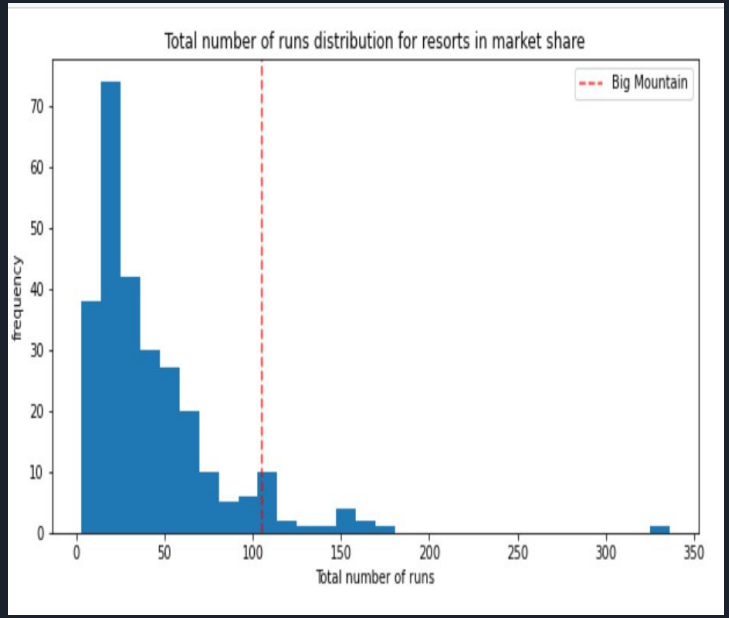
- Big Mountain has one of the longest runs compared to others
- Increase support for increase Adult Weekend Price

Vertical Drop (ft) Distribution for Resorts in Market Share

- Big Mountain has one of the highest vertical drop compared to others
- Increase support for price increase



Total Number of Runs Distribution for Resorts in Market Share



- Big Mountain has one of the higher total number of runs
- Supports increase pricing

Summary and Conclusion

- Modeled price = $95.87, actual price = $81.00, expected mean absolute error = $10.39
  - Resort is undercharging
- **Modified scenario 1**
  - Permanently closing down 1 of the least used runs
  - Makes no impact on revenue and ticket price
- **Scenario 2**
  - Installing additional chair lift will increase the vertical drop by 150 feet
  - Increases support for ticket price by $1.99 = expected to amount to $3,474,638
  - Cost of an additional chair increase will be $1,540,000 = net profit of around $2000000.
  - One of highest snow coverage and vertical drop, longest run, and total number of runs
- **I recommend charging $91**